Numerical Analysis I

Quiz 10A Name:_____

Consider the nonlinear model

$$y(t;c) = \frac{c}{1+ct}$$

and the two data points

$$(t_1, y_1) = (0, 2), \qquad (t_2, y_2) = (1, 1).$$

We wish to find the parameter c that minimizes

$$f(c) = \frac{1}{2}\sum_{i=1}^{2} \left(y(t_i;c) - y_i\right)^2 .$$

Starting from $c^{(0)} = 1$, perform one iteration of the Gauss–Newton method. Show your work and give the updated parameter $c^{(1)}$.

The residual vector is

$$r(c) = \begin{pmatrix} c-2 \\ \dfrac{c}{1+c} - 1 \end{pmatrix}.$$

The Jacobian is

$$J(c) = \begin{pmatrix} 1 \\ \dfrac{1}{(1+c)^2} \end{pmatrix}.$$

At $c^{(0)} = 1$, we have

$$r(1) = \begin{pmatrix} -1 \\ -\dfrac{1}{2} \end{pmatrix}, \qquad J(1) = \begin{pmatrix} 1 \\ \dfrac{1}{4} \end{pmatrix}.$$

The Gauss–Newton step p satisfies

$$J(1)^T J(1) p = -J(1)^T r(1).$$

Substituting,

$$\left(1 + \frac{1}{16}\right) p = 1 + \frac{1}{8},$$

so

$$\frac{17}{16} p = \frac{9}{8}, \qquad p = \frac{18}{17}.$$

Therefore, the updated parameter is

$$\boxed{c^{(1)} = c^{(0)} + p = 1 + \frac{18}{17} = \frac{35}{17}.}$$